

Analysis of Factors Influencing Content Discoverability on Instagram Using Machine Learning

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Abstract—Content discoverability is an important aspect of modern social media platforms because it determines how users encounter content beyond the accounts they directly follow. Instagram’s Explore page provides personalized recommendations intended to help users discover potentially relevant content. The exact mechanisms used by commercial recommendation systems are proprietary and depend on a large number of behavioral and content-related variables. Therefore, this study investigates a limited set of observable variables using a machinelearning-based approach.

The dataset used in this study contains 7,488 records and 9 columns, with important variables including user ID, photo ID, emoji usage, hashtag count, and comment text. Data preprocessing was performed by checking missing values, converting emoji usage into a binary representation, calculating comment length, and preparing textual information for sentiment analysis. Exploratory data analysis was then conducted to study user activity, hashtag usage, comment length, and content diversity. Sentiment analysis was performed using the TextBlob library.

For machine-learning analysis, emoji usage was selected as the target variable, while user ID, photo ID, comment information, and hashtag count were considered as input variables. A Random Forest Regressor was developed as the primary model and a single Decision Tree was used for comparison. Feature-importance analysis was performed to identify variables having greater influence within the model. The analysis indicates that hashtag-related variables show substantial importance, while comment length has comparatively less influence in the implemented model.

The findings provide an exploratory view of relationships between social-media interaction characteristics and content-related discoverability. However, the results cannot be interpreted as a reconstruction of Instagram’s actual Explore-page algorithm because the available dataset contains only a small subset of the variables used by a real-world recommendation system.

Index Terms—Instagram, Content Discoverability, Machine Learning, Social Media Analytics, Random Forest, Decision Tree, Sentiment Analysis, Hashtag Analysis, User Engagement

I. INTRODUCTION

Social-media platforms have become major sources of information, entertainment, communication, and content creation. The number of posts generated on these platforms is extremely large, making it impossible for users to manually examine all available content. Recommendation systems therefore play an important role in selecting and presenting content that may be relevant to individual users.

Instagram provides several mechanisms for content discovery, including its Explore page. The Explore page presents content that may be of interest to a user even when the user does not directly follow the account that created the content. This makes content discoverability an important factor for both users and content creators.

For users, effective recommendations can reduce information overload and help them discover content relevant to their interests. For content creators, greater discoverability can provide opportunities for increased engagement, audience growth, and recognition.

Understanding the factors associated with content discoverability is therefore an important research problem. However, commercial social-media recommendation systems are complex and proprietary. Their decisions may depend on many variables related to user behavior, content characteristics, interaction history, and contextual information.

The objective of this project is not to reproduce the complete Instagram recommendation algorithm. Instead, the study examines a limited set of observable variables from a supplied dataset and uses machine-learning techniques to identify patterns within the data.

The variables investigated include user identifiers, photo identifiers, emoji usage, hashtag counts, and comment text. In addition to machine learning, exploratory data analysis and sentiment analysis are used to obtain a broader understanding of the dataset.

The main objectives of the study are:

- 1) To study factors associated with content discoverability.
- 2) To analyze user activity and content diversity.
- 3) To investigate hashtag usage.
- 4) To analyze comment characteristics.
- 5) To perform sentiment analysis on user comments.
- 6) To develop a machine-learning model using the available variables.
- 7) To analyze feature importance.
- 8) To identify limitations of using a small dataset to study a large-scale recommendation problem.

II. OBJECTIVE AND MOTIVATION

The primary objective of this work is to investigate whether observable social-media interaction variables can provide useful information for understanding content discoverability.

The motivation for the study comes from the increasing importance of recommendation systems in social-media platforms. Users are continuously exposed to large quantities of content, while content creators compete for visibility.

The project considers the following research questions:

- 1) What observable characteristics are present in the available social-media dataset?
- 2) How are hashtags distributed among the observations?
- 3) What is the distribution of comment sentiment?
- 4) Is comment length strongly associated with the modeled interaction behavior?
- 5) Which available variables have greater importance in the machine-learning model?
- 6) Can a tree-based machine-learning model identify useful patterns from the available variables?

III. DATASET OVERVIEW

The dataset used for this project contains 7,488 rows and 9 columns. Important variables identified in the project include user ID, photo ID, emoji usage, hashtag count, and comments.

TABLE I
IMPORTANT VARIABLES USED IN THE STUDY

Variable	Description
User ID	Identifier associated with a user
Photo ID	Identifier associated with a content item
Emoji Usage	Indicator representing whether an emoji is used
Hashtag Count	Number of hashtags used
Comment	Textual content of a user comment
Comment Length	Length of the comment text
Sentiment	Sentiment obtained from comment analysis

The project report states that the dataset contains no missing values. Emoji usage was converted into a binary variable, and comment length was calculated as an additional feature.

IV. DATA PREPROCESSING

Data preprocessing is an important stage in a machine-learning pipeline because raw data may contain missing values, inconsistent representations, or variables that are unsuitable for direct modeling.

A. Missing Value Analysis

The dataset was examined for missing values. The project report indicates that no missing values were found. Consequently, no missing-value imputation was required.

B. Emoji Conversion

Emoji usage was converted into a binary representation. The transformation can be expressed as:

$$E = \begin{cases} 1, & \text{if an emoji is present,} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

C. Comment Length

Comment length was calculated from the comment text. For a comment c , its length can be represented as:

$$L(c) = |c| \quad (2)$$

where $|c|$ represents the number of characters in the comment.

D. Text Processing

Comment text was prepared for sentiment analysis. The project used TextBlob to determine the polarity of comments.

V. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was conducted to understand the structure and distribution of the available variables.

A. User ID Distribution

The project analysis indicates that most users fall within the 50–100 range of user identifiers, while relatively few observations are associated with values above 250.

It is important to distinguish an identifier from an activity measure. A larger User ID does not necessarily mean that a user is more active.

B. Hashtag Count

Hashtags are frequently used on social-media platforms to associate content with topics or categories.

The analysis indicates that the majority of comments contain one or two hashtags.

C. Comment Length

Comment length was investigated as a potential indicator of interaction depth.

Longer comments can contain more textual information than shorter comments. However, length alone does not necessarily indicate higher engagement or greater content visibility.

D. Unique Photo IDs

The number of unique Photo IDs associated with users was examined as an indicator of content diversity.

The project analysis reports that users associated with a higher number of unique Photo IDs were more likely to exhibit patterns associated with greater discoverability.

VI. SENTIMENT ANALYSIS

Sentiment analysis was performed to investigate the emotional tone of comments.

The project used the TextBlob Python library for this purpose.

For a comment c , TextBlob produces a polarity value $P(c)$.

The project categorizes comments according to their polarity:

$$S(c) = \begin{cases} \text{Positive,} & P(c) > 0, \\ \text{Neutral,} & P(c) = 0, \\ \text{Negative,} & P(c) < 0. \end{cases} \quad (3)$$

A. Sentiment Distribution

The project analysis reports that more than 99 percent of comments were classified as neutral.

Only a small fraction of comments were classified as positive or negative.

The strong concentration of observations in the neutral class means that sentiment has limited variation in this dataset.

VII. MACHINE LEARNING MODEL

The machine-learning component of the project investigates whether observable variables can be used to model emoji usage.

The project defines emoji usage as the target variable and uses comment information, user ID, photo ID, and hashtag count as input variables.

The general relationship can be represented as:

$$\hat{y} = f(X) \quad (4)$$

where X represents the input variables and $\hat{y}$ represents the predicted target.

The project uses a Random Forest Regressor as the primary model and a single Decision Tree for comparison.

A. Random Forest Regressor

Random Forest is an ensemble learning technique based on multiple decision trees.

For regression, individual trees generate predictions that are combined to obtain a final prediction.

The general prediction can be represented as:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(X) \quad (5)$$

where $T_i(X)$ represents the prediction of the i th tree and N represents the total number of trees.

B. Decision Tree

A single Decision Tree was developed as a comparison model.

A decision tree recursively divides observations according to feature-based decision rules.

Although a single tree is easier to interpret, it can be more sensitive to the training data than an ensemble of trees.

VIII. MODEL INPUTS AND TARGET

The project uses the following input variables:

- User ID
- Photo ID
- Comment information
- Hashtag count

The target variable is emoji usage.

The target is represented using the binary encoding described in the preprocessing section.

For a future predictive experiment, it is important to ensure that the target variable is not simultaneously included among the independent input variables. Such inclusion would result in target leakage and could produce misleadingly strong results.

IX. TRAINING AND TESTING

The project uses a machine-learning workflow in which the available observations are divided into training and testing sets.

The purpose of the training set is to allow the model to learn relationships between input variables and the target variable.

The testing set is used to evaluate the model on observations that were not used during training.

An 80:20 split can be represented as:

$$D = D_{train} \cup D_{test} \quad (6)$$

where approximately 80 percent of the observations are used for training and 20 percent are used for testing.

X. MODEL EVALUATION

Model evaluation is required to determine whether the trained model provides useful predictions on unseen observations.

For a regression formulation, common evaluation measures include Mean Squared Error (MSE) and Root Mean Squared Error (RMSE).

Mean Squared Error is defined as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (7)$$

Root Mean Squared Error is defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (8)$$

The numerical performance values were not included in the supplied project report. Therefore, numerical values are not invented in this paper.

TABLE II
QUALITATIVE FEATURE IMPORTANCE OBSERVED IN THE PROJECT

Feature	Observation
Hashtag Count	High importance in the implemented analysis
Emoji-related information	Important interaction signal
Comment Length	Comparatively limited importance
User ID	Used as an input variable
Photo ID	Used as an input variable

XI. FEATURE IMPORTANCE ANALYSIS

Feature-importance analysis was performed to understand which available variables contributed most strongly to the trained model.

The project identifies hashtag count and emoji-related information as important variables.

Comment length, in contrast, was reported to have comparatively limited influence.

Feature importance should not be interpreted as causal importance. A feature can be useful for prediction without directly causing the predicted behavior.

XII. RESULTS AND INSIGHTS

The analysis provides several observations about the supplied social-media dataset.

First, hashtag usage is concentrated around one or two hashtags for most observations.

Second, the sentiment analysis shows that more than 99 percent of comments are classified as neutral.

Third, the analysis of Photo IDs suggests that users associated with a greater variety of content exhibit patterns associated with higher discoverability.

Fourth, the feature-importance analysis indicates that hashtag count has greater importance than comment length in the implemented model.

These observations suggest that content-related and interaction variables can provide useful information for exploratory discoverability analysis.

However, they do not establish that these variables directly control Instagram’s Explore-page ranking.

XIII. DISCUSSION

The results indicate that machine-learning methods can be used to identify patterns within social-media interaction data.

The importance of hashtag count is particularly relevant because hashtags provide information about the topics associated with content.

The relatively low importance of comment length is also notable. The initial assumption that longer comments necessarily represent stronger engagement is not strongly supported by the feature analysis performed in this project.

The sentiment analysis provides an additional perspective on the comment data. However, because more than 99 percent

of comments are classified as neutral, sentiment has limited diversity in the available observations.

Content diversity also appears to be relevant. Users associated with multiple Photo IDs may represent users contributing a wider range of content.

Nevertheless, the relationship should be interpreted carefully. The available dataset does not contain direct measurements of Explore-page exposure, reach, or ranking position.

Therefore, the machine-learning model identifies patterns in observable interaction behavior rather than directly predicting the actual Instagram recommendation decision.

XIV. LIMITATIONS

Several limitations should be considered when interpreting the results.

A. Dataset Size

The dataset contains only 7,488 observations. This is small when compared with the scale of interactions processed by a major social-media platform.

Consequently, the results may not generalize to the complete Instagram population.

B. Limited Feature Set

The dataset contains only a small number of variables.

Real-world recommendation systems can use many additional variables, including interaction history, viewing behavior, shares, saves, timestamps, relationships between users, and other contextual information.

These variables are not available in the present dataset.

C. Proprietary Algorithm

Instagram’s recommendation mechanisms are proprietary.

The present study therefore cannot reproduce or verify the actual Explore-page algorithm.

The model should instead be viewed as an exploratory framework for analyzing observable variables.

D. Target Variable

Emoji usage is used as the target variable in the project.

Emoji usage is an observable interaction characteristic but is not a direct measurement of content discoverability.

A more rigorous future study should use direct measurements such as impressions, reach, Explore-page exposure, or ranking position.

E. Identifier Variables

User ID and Photo ID are identifiers.

Treating identifiers directly as numerical predictive variables may cause the model to learn dataset-specific patterns that do not generalize to other datasets.

Future experiments should replace identifiers with meaningful aggregated behavioral features where possible.

F. Sentiment Analysis

TextBlob provides a simple sentiment-analysis method.

The overwhelming neutral sentiment distribution suggests that more advanced NLP methods may be required to capture subtle semantic differences between comments.

XV. THREATS TO VALIDITY

The first threat to validity is sampling bias. The dataset may not represent the complete population of Instagram users or content.

The second threat is temporal validity. Social-media behavior and recommendation systems can change over time.

The third threat is measurement validity. Emoji usage is only an indirect interaction signal and cannot be treated as equivalent to content discoverability.

The fourth threat is the distinction between correlation and causation. A variable that is important to a predictive model does not necessarily cause the predicted outcome.

Finally, the limited number of available variables means that important factors influencing real-world content recommendation cannot be represented by the current model.

XVI. FUTURE WORK

The current project provides a foundation for several possible extensions.

A. Direct Discoverability Prediction

Future datasets should include direct discoverability measures such as:

- Explore-page impressions
- Content reach
- Saves
- Shares
- Click-through rate
- Engagement rate
- Follower growth

These variables would allow discoverability to be modeled directly rather than using emoji usage as a proxy.

B. Advanced NLP

Future work can use more advanced natural-language-processing techniques.

Possible approaches include TF-IDF representations, word embeddings, transformer-based representations, and pretrained language models.

These approaches could capture semantic information that is not available through simple comment-length or polarity measures.

C. Additional Machine-Learning Models

Future experiments can compare Random Forest and Decision Tree models with additional algorithms such as:

- Logistic Regression
- Support Vector Machines
- Gradient Boosting
- XGBoost

- LightGBM
- Neural Networks

Such comparisons would provide a more comprehensive evaluation of the predictive task.

D. Cross-Validation

Instead of relying on a single train-test split, future work should use cross-validation.

In k -fold cross-validation, the dataset is divided into k subsets. Each subset is used as a validation set while the remaining subsets are used for training.

This produces a more robust estimate of model performance.

E. Explainable Machine Learning

Explainable machine-learning techniques can be used to understand individual model predictions.

Methods such as SHAP can provide detailed information about how different variables contribute to predictions.

F. Temporal Analysis

Adding timestamps would allow the study to investigate how engagement and discoverability change over time.

Potential temporal features include posting time, time since publication, engagement growth, and recent user activity.

G. Larger Dataset

A substantially larger dataset would improve the statistical reliability and generalizability of the findings.

It would also allow the analysis to investigate more complex interactions between users, content, and engagement.

XVII. CONCLUSION

This paper presented an exploratory machine-learning-based analysis of factors associated with social-media content discoverability.

The study used a dataset containing 7,488 observations and 9 columns. The available variables included user ID, photo ID, emoji usage, hashtag count, and comment text.

The data-preprocessing stage involved checking missing values, converting emoji usage into a binary representation, calculating comment length, and preparing comments for sentiment analysis.

Exploratory analysis revealed that most observations contain one or two hashtags. The analysis also examined user distribution, comment length, and content diversity through unique Photo IDs.

Sentiment analysis using TextBlob showed that more than 99 percent of comments were classified as neutral. This highly concentrated distribution indicates that sentiment has limited variation in the available dataset.

For machine-learning analysis, emoji usage was selected as the target variable, while user ID, photo ID, comment information, and hashtag count were used as input variables. A Random Forest Regressor was developed as the primary model and a single Decision Tree was used for comparison.

The project analysis indicates that the Random Forest approach performed better than the single Decision Tree.

Feature-importance analysis further indicated that hashtag-related variables had substantial importance, while comment length had comparatively less influence.

The findings provide useful observations about the supplied dataset, but they should not be interpreted as a direct reconstruction of Instagram’s Explore-page algorithm. The real recommendation system is substantially more complex and may consider many variables unavailable in this dataset.

The most important direction for future work is therefore to replace the proxy target with direct discoverability measures such as impressions, reach, saves, shares, and Explore-page exposure.

Larger datasets, advanced NLP techniques, additional machine-learning algorithms, cross-validation, temporal analysis, and explainable AI could further improve the proposed framework.

Overall, the study demonstrates that machine-learning techniques can provide a useful exploratory framework for studying social-media interaction patterns while emphasizing the importance of careful target definition, adequate data, and cautious interpretation of predictive relationships.

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